

df tensor, symmetric, exterior algebras

Vincent Wang

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1 Tensor Algebras

Define $\mathcal{T}^k(M) = M \otimes_R M \otimes_R \cdots \otimes_R M$ and $\mathcal{T}^0(M) = R$. We have $\mathcal{T}^k(M) \otimes \mathcal{T}^j(M) = \mathcal{T}^{k+j}(M)$. Define $\mathcal{T}(M) = \bigoplus_{k \geq 0} \mathcal{T}^k(M)$.

Claim: Suppose $\varphi : M \rightarrow A$ is an R -module homomorphism. Then φ extends to an R -algebra homomorphism $\Phi : \mathcal{T}(M) \rightarrow A$.

Proof: We wish for $\Phi(m_1 \otimes m_2 \otimes \cdots \otimes m_k) = \Phi(m_1)\Phi(m_2) \cdots \Phi(m_k) = \varphi(m_1)\varphi(m_2) \cdots \varphi(m_k)$. So we first have the map $\varphi_k : M^k \rightarrow A$ given by $\varphi_k(m_1, m_2, \dots, m_k) = \varphi(m_1)\varphi(m_2) \cdots \varphi(m_k)$, which is R -linear in each argument, so we descend to $\psi_k : M \otimes_R M \otimes_R \cdots \otimes_R M \rightarrow A$, where the tensor is taken k times. By direct sum properties, we have the desired $\Phi : \mathcal{T}(M) \rightarrow A$.

$\mathcal{T}(M)$ is called the tensor algebra of M .

Claim: $M = (\mathbb{Q}/\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Q}/\mathbb{Z}) = 0$ (divisible and torsion).

Proof: We have

$$\overline{a/b} \otimes \overline{c/d} = \overline{a} \otimes \overline{c/(bd)} = 0 \otimes \overline{c/(bd)} = 0.$$

Then $\mathcal{T}(\mathbb{Q}/\mathbb{Z}) = \mathbb{Z} \oplus (\mathbb{Q}/\mathbb{Z})$. Multiplication: $(a, \bar{p})(b, \bar{q}) = (ab, \overline{aq + bp})$. $\mathcal{T}(\mathbb{Z}/n\mathbb{Z}) = \mathbb{Z} \oplus (\mathbb{Z}/n\mathbb{Z}) \oplus (\mathbb{Z}/n\mathbb{Z}) \oplus \cdots$, so $\mathcal{T}(\mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}[x]/\langle nx \rangle$.

Let $\mathcal{C}(M)$ denote the ideal generated by elements $m_1 \otimes m_2 - m_2 \otimes m_1$ for $m_1, m_2 \in M$ and $\mathcal{S}(M) = \mathcal{T}(M)/\mathcal{C}(M)$.

1. Let $M = \langle x_0 \rangle$. $\mathcal{C}(M)$ is the ideal generated by

$$rx_0 \otimes sx_0 - sx_0 \otimes rx_0 = (rs - sr)(x_0 \otimes x_0) = 0,$$

since R is commutative. Thus, $\mathcal{S}(M) = \mathcal{T}(M)/\mathcal{C}(M) = \mathcal{T}(M)$.

7. Let $R = \mathbb{Z}[x, y]$ and $I = (x, y)$.

(a). Suppose $ax + by = a'x + b'y$, so $(a - a')x = (b' - b)y$, so $y \mid (a - a')$. Write $a - a' = yf$.

Then $xyf = (b' - b)y$, so $b' - b = xf$. Thus, $(a', b') = (a - yf, b + xf)$.

(b). Prove that $\varphi(ax + by, cx + dy) = (ad - bc) \pmod{(x, y)}$ is a well defined alternating bilinear map $I \times I \rightarrow \mathbb{Z} = R/I$. Suppose $a'x + b'y = ax + by, c'x + d'y = cx + dy$. Let f, g be such that $a' = a + yf, b' = b - xf, c' = c + yg, d' = d - xg$. Then

$$\begin{aligned} a'd' - b'c' &= (a + yf)(d - xg) - (b - xf)(c + yg) \\ &= (ad - bc) + xh + yk \end{aligned}$$

for some $h, k \in R$. We check that $\varphi(ax + by, ax + by) = 0$ and is bilinear. As a bilinear map, $\varphi \neq 0$, so $\bigwedge^2 I \neq 0$ while $\bigwedge^2 R = 0$, so if $M \hookrightarrow N$, we need not have $\bigwedge^k(M) \hookrightarrow \bigwedge^k(N)$.

13. Let F be a field with $\text{char } F \neq 2$. Then $\mathcal{T}^2(V) = \mathcal{S}^2(V) \oplus \bigwedge^2(V)$.

Proof: Suppose $v \otimes w \in \mathcal{T}^2(V)$. Let $x = (v + w)/2$ and $y = (v - w)/2$. Then

$$\begin{aligned} v \otimes w &= (x + y) \otimes (x - y) = x \otimes x + y \otimes x - x \otimes y - y \otimes y \\ &= (x \otimes x - y \otimes y) + (y \otimes x - x \otimes y), \end{aligned}$$

so $\mathcal{T}^2(V)$ is the sum of $\mathcal{S}^2(V), \bigwedge^2(V)$. Suppose $z \in \mathcal{S}^2(V) \cap \bigwedge^2(V)$. Then $(12)z = z = -z$, so $z = 0$.

Claim: Let R be such that $k! \in R^\times$. Then $\text{Alt}^k(M) \cong \bigwedge^k(M)$.

Proof: Define $\pi : \mathcal{T}^k(M) \rightarrow \text{Alt}^k(M)$ by $\pi(z) = \sum_{\sigma \in S_k} \epsilon(\sigma)\sigma z$. π is well-defined since

$$\tau\pi(z) = \sum_{\sigma \in S_k} \epsilon(\sigma)\tau\sigma(z) = \epsilon(\tau) \sum_{\sigma' \in S_k} \epsilon(\sigma')\sigma'(z) = \epsilon(\tau)\pi(z),$$

so $\pi(z)$ is indeed an alternating tensor. Since $k!$ is invertible, let $\pi' = \frac{\pi}{k!}$, so π' fixes $\text{Alt}^k(M)$ and is surjective. I claim that $\ker \pi' = \mathcal{A}^k(M)$. Consider $z = m_1 \otimes m_2 \otimes \cdots \otimes m_{i-1} \otimes m_i \otimes m_i \otimes m_{i+2} \otimes \cdots \otimes m_k$. Write

$$\begin{aligned} \pi'(z) &= \frac{1}{k!} \left(\sum_{\sigma \in A_k} \epsilon(\sigma)\sigma z + \sum_{\sigma \in A_k} \epsilon(\sigma(i, i+1))\sigma(i, i+1)z \right) \\ &= \frac{1}{k!} \left(\sum_{\sigma \in A_k} \sigma z - \sum_{\sigma \in A_k} \sigma z \right) = 0, \end{aligned}$$

so $\mathcal{A}^k(M) \leq \ker \pi'$. Now suppose $\pi'(z) = 0$. In general, we have:

$$z - \pi'(z) = \frac{1}{k!} \sum_{\sigma \in S_k} (z - \epsilon(\sigma)\sigma z)$$

The RHS lies in $\mathcal{A}^k(M)$ since every permutation is a product of adjacent transpositions. For $\pi'(z) = 0$, we have $z \in \mathcal{A}^k(M)$. Thus, $\ker \pi' = \mathcal{A}^k(M)$ and $\bigwedge^k(M) \cong \text{Alt}^k(M)$.

Let A, B be $\mathbb{Z}$ -graded rings. Define $A \hat{\otimes} B$ to be the ring with grading $(A \hat{\otimes} B)_n = \bigoplus_{p+q=n} A_p \otimes B_q$ and multiplication on homogenous elements a, b, c, d by $(a \otimes b)(c \otimes d) = (-1)^{|b||c|} ac \otimes bd$. We have

$$|(-1)^{|b||c|} ac \otimes bd| = |ac| + |bd| = |a \otimes b| + |c \otimes d|,$$

so $(A \hat{\otimes} B)_m (A \hat{\otimes} B)_n \leq (A \hat{\otimes} B)_{m+n}$.

Claim: Let R be a $\mathbb{Z}$ -graded ring (not necessarily commutative) and suppose we have a bilinear map $\varphi : A \times B \rightarrow R$ such that $\varphi(A_m \times B_n) \leq R_{m+n}$. Then there is an induced homomorphism of graded rings $\Phi : A \hat{\otimes} B \rightarrow R$.

Claim: $\bigwedge(M \oplus N) = \bigwedge M \hat{\otimes} \bigwedge N$ (as rings).

Proof: Define $\varphi : \bigwedge M \times \bigwedge N \rightarrow \bigwedge(M \oplus N)$ (as rings